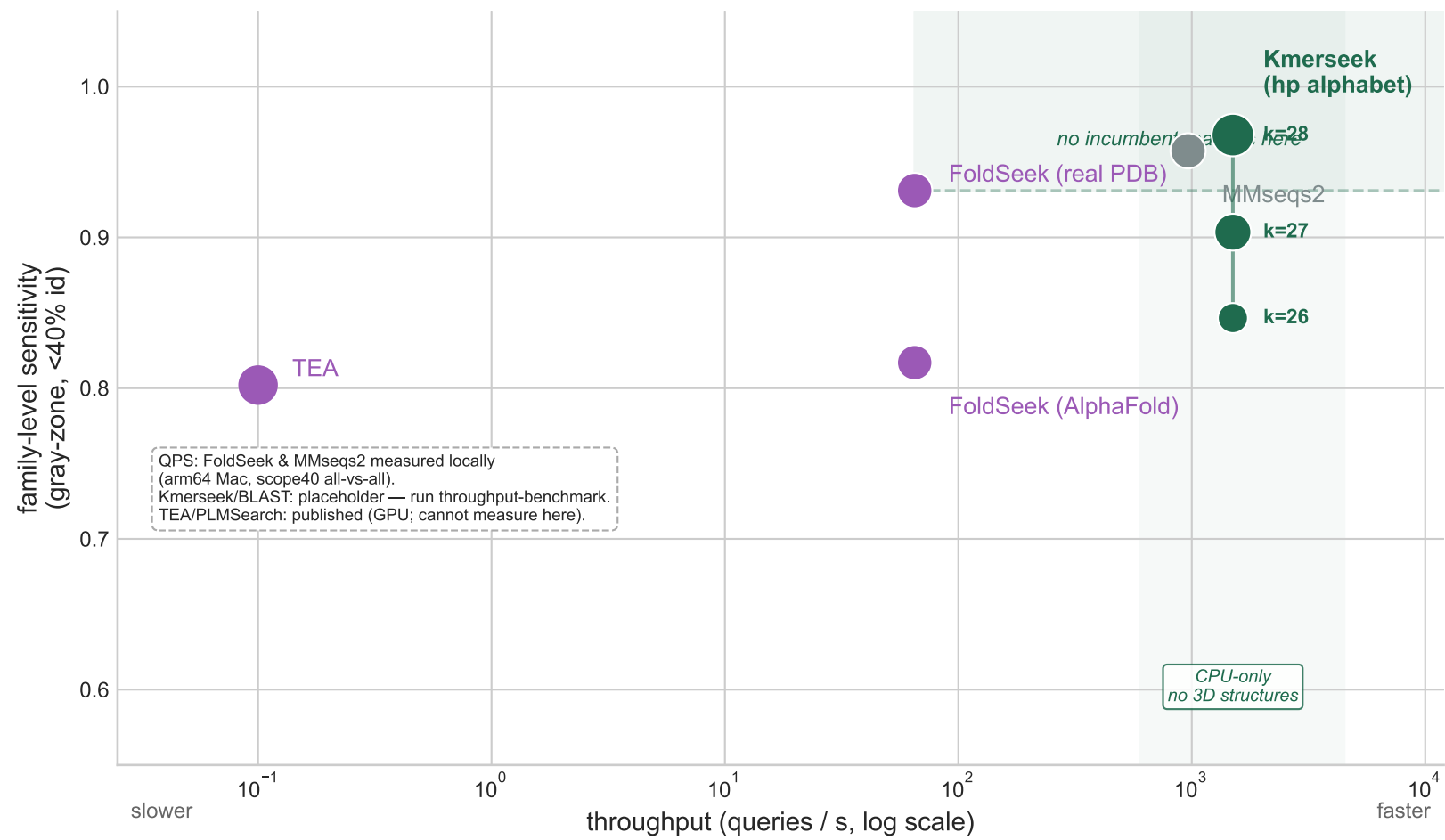


The frontier Kmerseek opens



Kmerseek (this work) needs 3D structure needs language model alignment				
Method	Needs 3D?	Alignment-free?	Fam AUC (gray-zone)	Hardware
Kmerseek k=26	No	Yes	~0.85	CPU only, no GPU
Kmerseek k=27	No	Yes	~0.90	CPU only, no GPU
Kmerseek k=28	No	Yes	~0.97	CPU only, no GPU
FoldSeek (PDB)	Yes	No (3D aln)	~0.93	CPU; pre-built crystal structure DB
FoldSeek (AlphaFold)	Yes	No (3D aln)	~0.82	CPU search; GPU for AF2 prediction
TEA	Yes	No (embed)	~0.80	CUDA GPU required (ESMFold)
PLMSearch	No ¹	No (embed)	—	CUDA GPU required (ESM2)
MMseqs2 / BLAST	No	No (seq aln)	— (low)	CPU only, no GPU

¹ PLMSearch uses a protein language model.
FoldSeek & MMseqs2 QPS measured locally on arm64 Mac (scope40 all-vs-all, no pre-built DB). Kmerseek & BLAST QPS are placeholders — run nextflow-runs/throughput-benchmark to measure (also provides peak RAM and CPU via Nextflow trace). TEA & PLMSearch QPS from published papers (NVIDIA CUDA GPU; cannot run on Apple Silicon). AUC = sensitivity-to-first-FP, SCOPe40 gray-zone queries (<40% id).